

FLS Realized-Geography Composition Analog

Scope

The public data do not identify NASS's confidential operation-level FLS sample, response adjustments, or survey weights. This procedure therefore constructs:

- a soft-calibrated central OEWS-area geographic distribution;
- a reproducible ensemble of plausible realized sampling variation; and
- descriptive simulation envelopes for that predeclared ensemble.

It does not claim to recover the original survey weights. Estimation uses no FLS wage, OEWS wage, AEWR policy value, first-stage estimate, or outcome. The supported source years are 2011–2021, and every FLS region-year is estimated independently.

The implementation is:

```
uv run python code/c03_iv/10_recover_fls_realized_geography.py
uv run python code/c03_iv/12_validate_fls_realized_geography_wages.py
```

The second command is a held-out descriptive check run only after the weight artifacts and specification labels have been finalized.

Expected geographic prior

Let M_{ct} be the county mass from `weight_spec = "census_hired_workers_qcew_updated"`. It anchors 2007, 2012, 2017, and 2022 to Census of Agriculture directly hired workers, updates annual employment with strict QCEW, QWI, and BEA sources, applies a two-sided path correction, and rakes county mass to the interpolated state Census total.

Let h_{cit} be OEWS area i 's share of distinct township codes mapped within county c , where

$$\sum_i h_{cit} = 1.$$

The allocated area mass is

$$A_{irt} = \sum_{c \in r} h_{cit} M_{ct}.$$

An OEWS area is identified by the pair (FLS region, OEWS area code), so an area spanning regions remains separated. The expected geographic prior is

$$q_{irt} = \frac{A_{irt}}{\sum_{j \in r} A_{jrt}}$$

for positive area mass. Zero-mass areas remain outside the supported frame. Wage availability and OEWS employment never determine this support or prior. The county allocation is retained so an area draw can later be expanded with the fixed baseline conditional county mass.

FLS joint targets

For each observed reference quarter q , the target has two duration cells:

- workers employed 150 days or more (**long**); and
- workers employed fewer than 150 days (**short**).

Let N_{rtqd} denote the published count. The normalized target is

$$\tau_{rtqd} = \frac{N_{rtqd}}{\sum_{q,d} N_{rtqd}}.$$

The denominator is the sum of the long and short counts. Differences from the separately published all-worker count are retained as rounding/source diagnostics.

Targets are never imputed. A region-year requires at least three usable quarters and a positive duration-cell total. April duration counts are absent in 2011, so that year uses January, July, and October: six joint cells and five independent contrasts. Each year from 2012 through 2021 uses four quarters: eight joint cells and seven independent contrasts.

Public area features

Seasonal employment intensity

For each county-year-quarter, private QCEW NAICS 111 and 112 employment is valid only when both industries are disclosed. If that strict QCEW cell is incomplete, private QWI beginning-of-quarter employment fills it when both industries are available.

County employment is allocated with the same township shares used for the frame prior. An area-quarter aggregate is usable only when its observed county inputs cover at least 90 percent of the area’s frame mass. Partial totals below that threshold remain missing.

Let E_{itq} be the accepted area-quarter employment and A_{it} the annual area frame mass. First form

$$I_{itq} = \frac{E_{itq}}{A_{it}}.$$

Over the quarters observed in the corresponding FLS target, define

$$S_{itq} = \frac{I_{itq}}{\sum_{q' \in Q_{rt}} I_{itq'}}.$$

Duration analog

QWI persistence is stable employment divided by beginning-of-quarter employment. In matched Census benchmark area-quarter cells, estimate the frame-weighted median odds ratio

$$R = \text{weighted median} \left[\frac{\text{odds(QWI persistence)}}{\text{odds(Census 150-plus-day share)}} \right].$$

The implementation estimates one public bridge from matched cells and records the number of matched area-quarter and area-year cells. It uses no FLS data. Translate QWI persistence to the long-duration analog:

$$D_{itq} = \text{logistic} \left[\text{logit}(\text{QWI persistence}_{itq}) - \log R \right].$$

When QWI duration is unavailable, use the area's Census 150-plus-day share, interpolated between benchmark years on the logit scale.

Public-only imputation hierarchy

Any remaining public covariate gap follows this fixed hierarchy:

1. same-area, same-quarter linear interpolation over time, using the log scale for employment intensity and the logit scale for duration;
2. the nearest available same-area, same-quarter value within two years;
3. the frame-prior-weighted FLS-region-year mean for the quarter; and
4. failure of the region-year if no regional value exists.

The artifact records the source, coverage, and imputation flag for each feature, plus employment, duration, and combined imputed prior mass. This borrowing fills public covariates only; it never smooths recovered weights across years.

Joint compositions and Helmert moments

Each area has the joint composition

$$X_{itq,\text{long}} = S_{itq} D_{itq},$$

$$X_{itq,\text{short}} = S_{itq}(1 - D_{itq}).$$

The cells are normalized to sum to one for every supported area. Both the area compositions and FLS target are projected onto a $K \times (K - 1)$ orthonormal Helmert basis. This avoids choosing an omitted category.

For contrast j , let Z_{ij} be the area contrast and T_j the target contrast. Standardize using the deterministic frame prior:

$$\begin{aligned}\bar{Z}_j^q &= \sum_i q_i Z_{ij}, \\ s_j^q &= \sqrt{\sum_i q_i (Z_{ij} - \bar{Z}_j^q)^2}, \\ \tilde{Z}_{ij} &= \frac{Z_{ij} - \bar{Z}_j^q}{s_j^q}, \quad \tilde{T}_j = \frac{T_j - \bar{Z}_j^q}{s_j^q}.\end{aligned}$$

Only contrasts with effectively zero cross-area variation are dropped, and each drop is flagged.

Soft entropy recovery

For a prior vector p , solve

$$\min_{\mathbf{w}} D_{KL}(\mathbf{w} \| \mathbf{p}) + \frac{\rho}{2} \|\tilde{\mathbf{Z}}' \mathbf{w} - \tilde{\mathbf{T}}\|_2^2$$

subject to

$$w_i \geq 0, \quad \sum_i w_i = 1.$$

The dual has the exponential-tilting weights

$$w_i(\lambda) = \frac{p_i \exp(\tilde{\mathbf{Z}}'_i \lambda)}{\sum_k p_k \exp(\tilde{\mathbf{Z}}'_k \lambda)}$$

and objective

$$\log \left[\sum_i p_i \exp(\tilde{\mathbf{Z}}'_i \lambda) \right] - \lambda' \tilde{\mathbf{T}} + \frac{1}{2\rho} \lambda' \lambda.$$

The implementation uses a damped Newton solver with log-sum-exp evaluation. There is no exact wage constraint. Larger ρ pursues the FLS quarter-duration moments more strongly. The fixed values are

$$\rho \in \{0.01, 0.03, 0.10, 0.30, 1.00\},$$

with $\rho = 0.10$ primary. A deterministic center is solved from the original $\mathbf{q}$ for every rho.

Dirichlet ensembles

For each region-year,

$$N_{rt}^{eff} = \frac{1}{\sum_i q_{irt}^2},$$

$$\kappa_{rt} = m N_{rt}^{eff},$$

and

$$\mathbf{q}_{rt}^{(b)} \sim \text{Dirichlet}(\kappa_{rt} \mathbf{q}_{rt}).$$

The fixed dispersion multipliers are

$$m \in \{2, 5, 10, 20\},$$

with $m = 10$ primary. Each prior draw is passed through the same soft entropy solver.

The primary specification is `fls_geo_dirichlet_m10_rho010` with 999 draws. Every non-primary rho at $m = 10$ has 199 common draws, and every non-primary m at $\rho = 0.10$ has 199 draws. The base seed is `20260726`, with deterministic offsets by region, year, and multiplier. Rho is omitted from the prior-draw seed so its sensitivity path uses identical priors. Partial and repeated builds therefore reproduce the same draw IDs and weights.

These dispersion and penalty values are declared sensitivity parameters. They are not estimated sampling variances and are never selected using wage fit, first-stage strength, or outcomes.

Outputs

The recovery produces:

- `fls_realized_geography_features.parquet`;
- `fls_realized_geography_county_area_prior.parquet`;
- `fls_realized_geography_area_prior.parquet`;
- `fls_realized_geography_draws/`, partitioned by region, source year, and specification;

- `fls_realized_geography_weight_summary.parquet`;
- `fls_realized_geography_diagnostics.parquet`; and
- `target`, `duration-bridge`, and feature diagnostic artifacts.

The summary contains the frame prior, deterministic center, draw mean, standard deviation, and 2.5th, 50th, and 97.5th simulation-envelope percentiles. The draw-level diagnostics contain standardized moment imbalance, KL divergence, effective area count, maximum area share, imputed prior mass, and solver status.

Every realized output identifies

- `weight_spec = "fls_realized_geography_dirichlet_entropy"`;
- `baseline_weight_spec = "census_hired_workers_qcew_updated"`;
- `moment_spec = "fls_joint_quarter_duration"`;
- `wage_target_used = FALSE`;
- `rho`;
- `kappa_multiplier`;
- `weight_draw_id`;
- `is_primary`; and
- `simulation_seed`.

If county weights are needed later, the fixed expansion is

$$w_{crt}^{(b)} = w_{irt}^{(b)} \frac{A_{cit}}{A_{irt}}.$$

No additional within-area county randomness is estimated.

Use in the dissimilarity IV

The IV construction uses only the deterministic center from the predeclared primary specification, `fls_geo_dirichlet_m10_rho010`. For each target cluster, the area center is expanded to counties with the fixed conditional shares above, restricted to the two selected donor clusters, and then re-aggregated to unique OEWS areas. Any OEWS area touching the target cluster is excluded. The remaining weights are normalized over areas with an observed OEWS agricultural wage.

The Census-frame version is retained as a benchmark on identical donor and wage support. Dirichlet draws and sensitivity specifications remain recovery diagnostics; they do not generate draw-specific instruments. The IV therefore uses FLS worker-composition moments but never an FLS wage target.

Held-out wage validation

Validation begins only after the recovery artifacts exist. Covered agricultural occupations are combined within each OEWS area using OEWS employment.

For a center or draw, weights are renormalized over the wage-observed area mass solely for the validation calculation.

The validation artifacts report predicted regional wages, residuals, MAE, RMSE, correlation, and simulation-envelope coverage. Results are descriptive and cannot alter the primary specification or sensitivity grid.

Acceptance checks

The implementation checks:

1. area priors, deterministic centers, and successful draws sum to one within 10^{-10} ;
2. target and area joint-cell vectors sum to one;
3. the 2011 and later-year cell/contrast counts follow the published target availability;
4. township allocation conserves county frame mass;
5. fixed county expansion reproduces every area weight;
6. every positive-prior area survives public-feature imputation;
7. calibration does not increase the targeted standardized-imbalance norm;
8. increasing rho weakly reduces that norm for common prior draws;
9. seeds, draw IDs, and weights are reproducible in partial builds;
10. the recovery runs without wage, policy, first-stage, or outcome artifacts;
11. every primary deterministic center succeeds; and
12. at least 99 percent of primary draws succeed in every region-year.

The FLS panel structure and confidential dual-frame design remain unidentified. Region-years are treated independently, and later Census benchmarks can revise the expected frame consistently with its two-sided construction.